

# Repaired Native-VC Brill Authority Qualification

AthenaK evidence campaign

24 August 2026

**NATIVE\_AMR\_UNSTABLE**

## Abstract

We generated a fresh N256 vertex-centered Cartoon AMR authority after repair of derefinement-slot corruption, verified deterministic record/replay, and replayed the authority at N128 and N512. The repaired scheme converges cleanly through central proper time  $\tau \simeq 4$ . A later N256 extension develops large constraints on a fixed 44-leaf hierarchy before native refinement activates, then undergoes a refinement cascade to logical level 23 and stalls near  $t = 11.192887945M$ . The result qualifies the early window but is neither a full Figure-3 reproduction nor a convergence result for late collapse.

## 1 Authority and deterministic replay

The reviewed numerical baseline is commit 6dd20656a305f2543bbbd7001550c6ac67019180. A diagnostics-only repair at d63519328214a6315a9cc1f7d5e4a1aa4bca21b0 replaces unordered CUDA atomics in regional history sums with a Kokkos reduction. It changes no evolution variable, AMR decision, transfer operator, equation, gauge, RK, KO, or CFL choice. In the post-repair aggregate host run, 140/141 enabled tests passed; the sole AMR-history timeout passed in isolation in 177.96 seconds. The focused one-A100 suite passed 20/20 tests.

N256 record and replay have identical accepted events, times, checksums, history bytes, auxiliary diagnostics, and numerical payloads for all 33 binary outputs and six restarts. Complete binary/restart containers differ in their authenticated parameter headers because one records mode **record** and the other **replay**; numerical identity follows the repository contract and hashes bytes after **<par\_end>**.

The repaired and historical authorities agree exactly through event 3. The historical tree immediately re-refines at  $t = 0.30305969M$  to 74 leaves. The repaired tree instead stays at 44 leaves through  $t = 6.5M$  and next refines only at  $t = 10.27866870M$  (Figure 1).

## 2 Qualified early and $\tau \simeq 4$ windows

N128, N256, and N512 replay the same logical tree at identical physical event times. At  $t = 6.5M$ , their central proper times are 3.98615, 3.98694, and 3.98712. Every global constraint decreases monotonically with resolution.

Table 1: Global constraint diagnostics at  $t = 6.5M$ .

| Resolution | $C$                     | $H$                     | $M$                     | $Z$                     |
|------------|-------------------------|-------------------------|-------------------------|-------------------------|
| N128       | $5.5801 \times 10^{-2}$ | $2.1558 \times 10^{-2}$ | $8.6428 \times 10^{-3}$ | $4.0712 \times 10^{-3}$ |
| N256       | $3.2850 \times 10^{-3}$ | $3.5072 \times 10^{-4}$ | $4.2914 \times 10^{-4}$ | $5.0544 \times 10^{-4}$ |
| N512       | $1.8080 \times 10^{-3}$ | $1.2559 \times 10^{-4}$ | $5.7452 \times 10^{-5}$ | $4.0103 \times 10^{-4}$ |

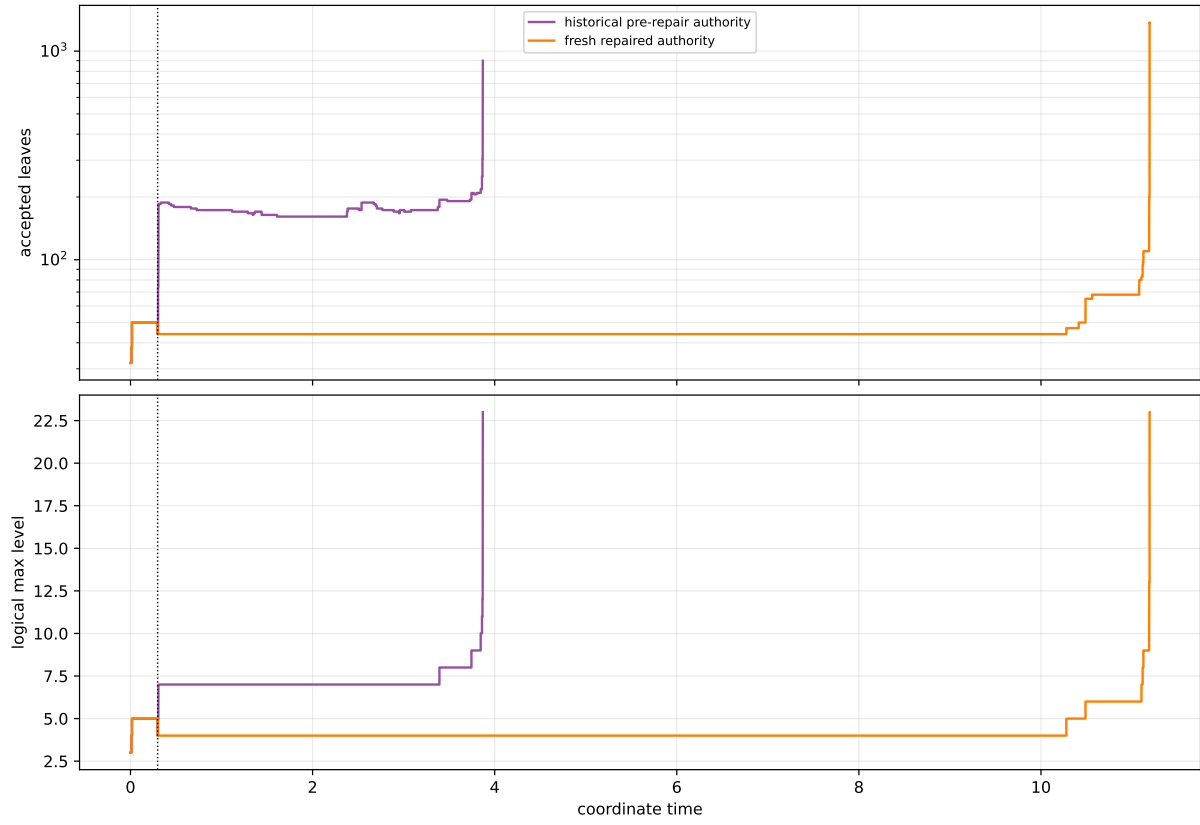


Figure 1: Historical pre-repair and fresh repaired hierarchy histories. The dotted line marks the repaired event-3 derefinement.

Median effective history orders are 6.746, 6.794, 6.127, and 7.029 for  $C, H, M, Z$ . Every nonzero evolved field has positive minimum trusted-core order; the lowest is 1.053 for  $\Theta$ . Minimum  $\chi$  is 0.3211, the minimum conformal-metric SPD pivot is 0.1974, and sampled same-level and coarse-fine shared-node spreads are zero.

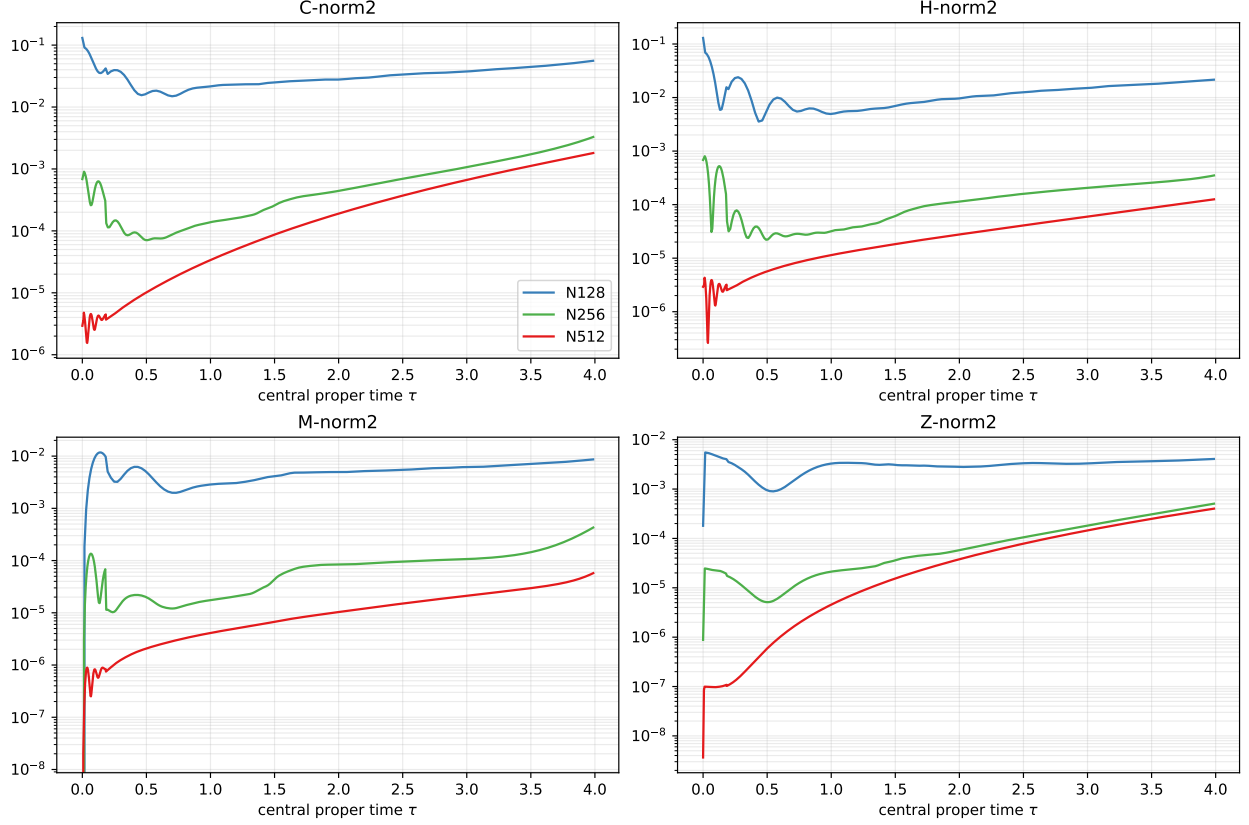


Figure 2: Global constraints through the qualified  $\tau \simeq 4$  milestone.

### 3 Native-AMR failure before $\tau \simeq 7$

The later N256 segment first exceeds  $C = 0.01$  at  $t = 7.2894893M$  ( $\tau = 4.45109$ ) and  $C = 1$  at  $t = 9.2042653M$  ( $\tau = 5.61265$ ), while the hierarchy remains fixed at 44 leaves and physical level 1. The first late refinement occurs only at  $t = 10.2786687M$  ( $\tau \simeq 6.28$ ), when the nearest history sample already has  $C \simeq 20.9$ .

The hierarchy then reaches logical level 7 at  $t = 11.1038229M$ . Event 22 at  $t = 11.19248494M$  reaches logical level 14 and begins an effectively coordinate-time-stagnant cascade. At cancellation, event 159 has 1,367 leaves, logical level 23, physical refinement level 20, and  $\Delta t = 1.07 \times 10^{-8}M$ . The global  $C$  diagnostic is  $1.40 \times 10^{14}$  and the domain maximum absolute Kretschmann scalar is  $1.60 \times 10^{25}$ , while the sampled axis-center value is only  $7.40 \times 10^{-3}$ . The runaway is therefore noncentral in this diagnostic.

### 4 Interpretation and qualification boundary

Observation: substantial constraint growth begins on the fixed parent hierarchy before the refinement cascade. Inference: the current chi-only sensor activates late relative to the shortest or error-bearing scale, or a bulk numerical mode grows before refinement exposes it. Hypothesis: repeated high-order

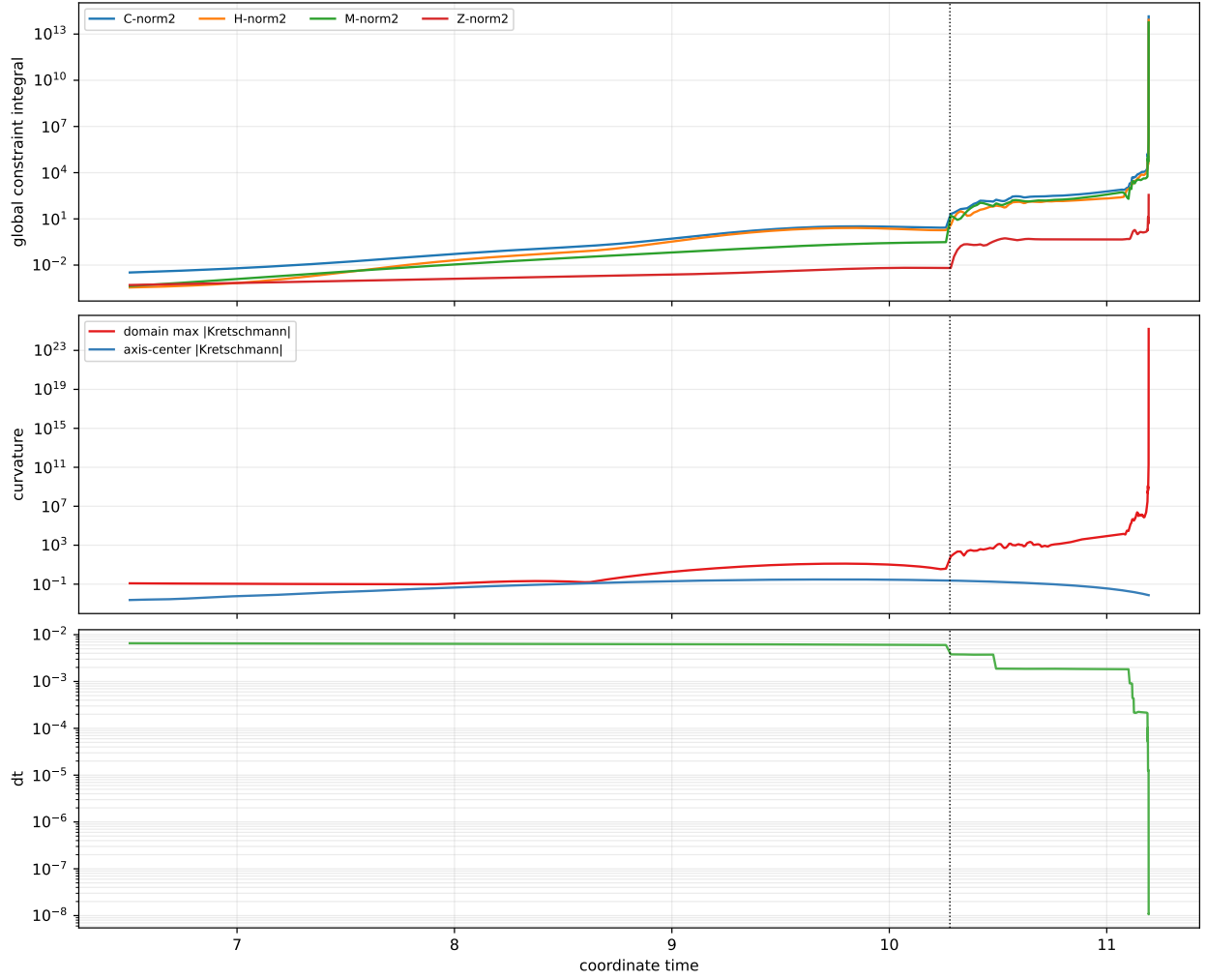


Figure 3: Constraint growth, noncentral curvature runaway, and timestep collapse in the failed N256 extension. The dotted line marks the first late accepted refinement.

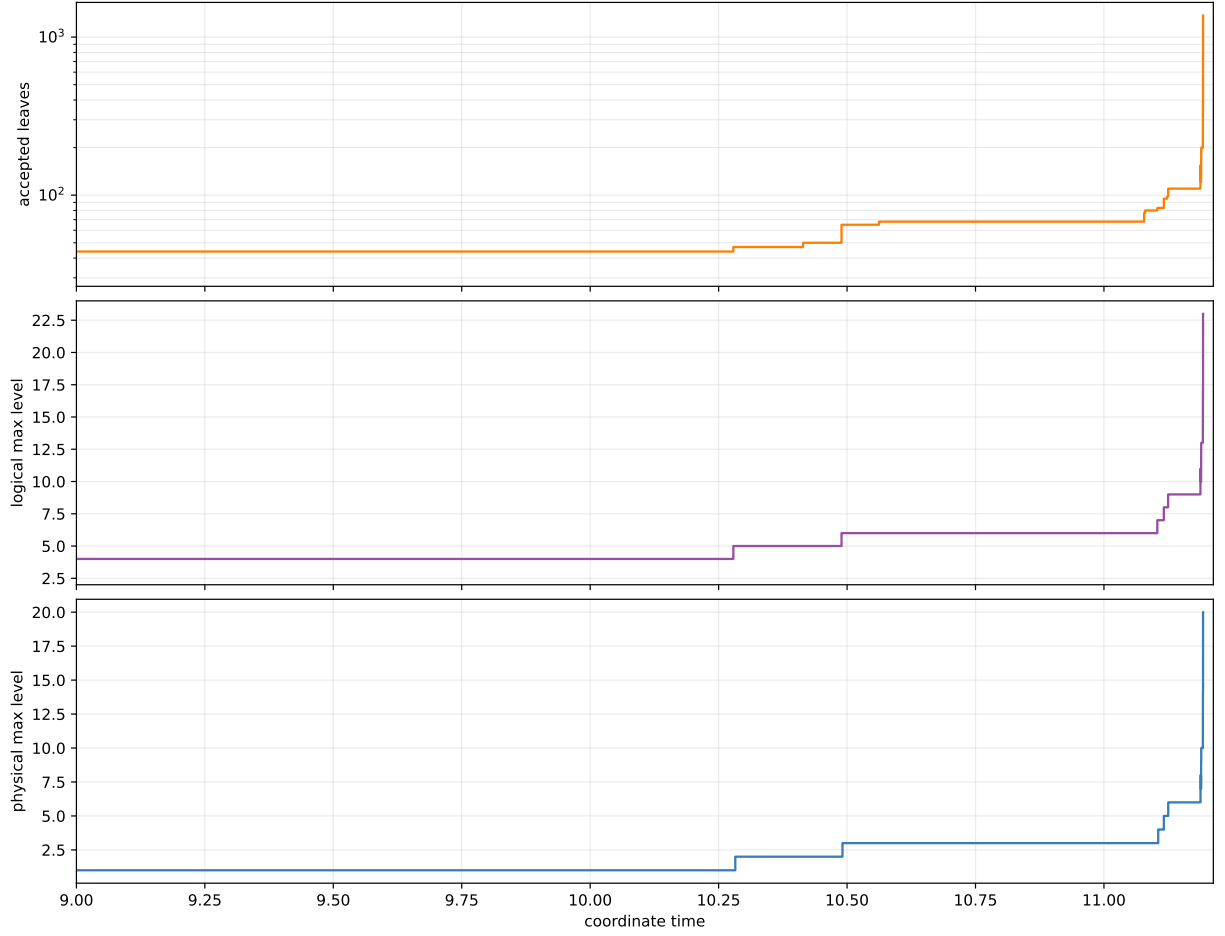


Figure 4: Accepted leaves and logical/physical refinement levels during the late native-AMR cascade.

transfer and interface coupling then accelerates the existing error. This campaign does not distinguish those mechanisms and does not establish a source-level bug.

Job 57525753 was cancelled at the fail gate. No N128/N512 replay was launched for the failed authority, and no  $\tau \simeq 10.5$  or full published-interval calculation was attempted. Figure 5 is only a direct, unshifted overlay over the qualified partial interval.

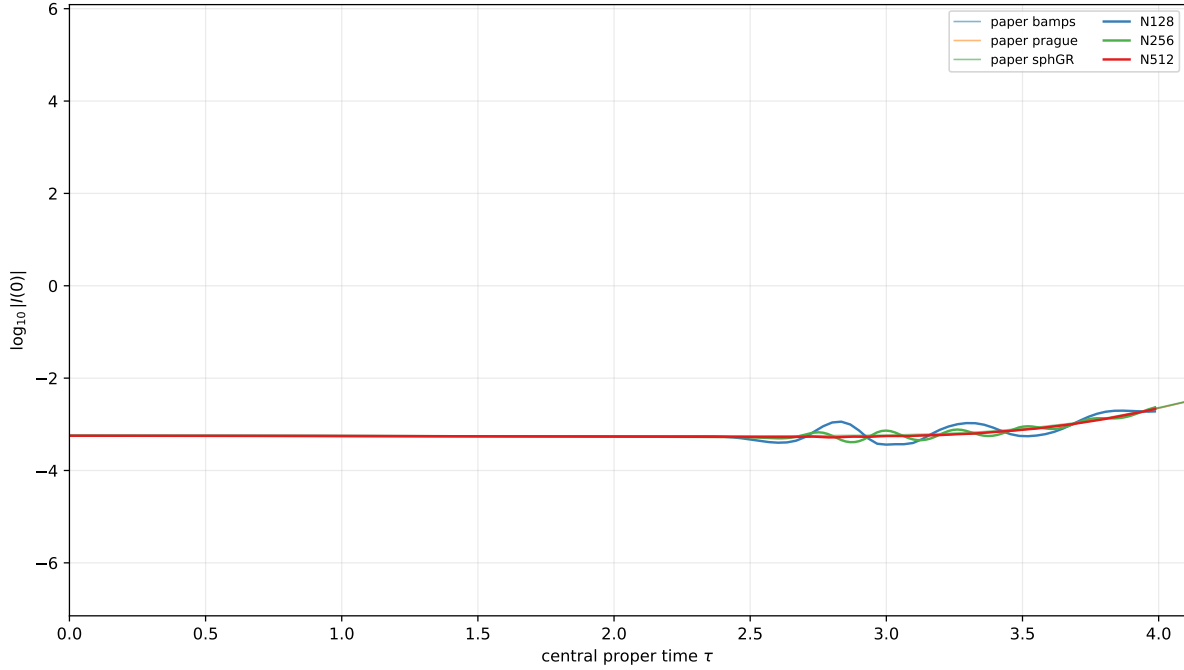


Figure 5: Authenticated Figure-3 curves and AthenaK results over the qualified partial interval. No time shift, amplitude scaling, or coordinate rescaling is applied.